

Commercial Wind Energy Installations and Local Economic Development: Evidence from U.S. Counties

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Abstract

We examine the causal impact of wind energy installation on the local economies of counties in the United States. Using data on the universe of commercial wind energy installations from 1995-2018 and a difference-in-differences instrumental variable identification strategy, we find that wind energy installation led to economically meaningful and statistically significant increases in county GDP per-capita, income per-capita, median household income, and median home values. We also find evidence that while wind energy installation has little effect on total employment, the composition of local employment shifts away from farm towards non-farm employment, notably leading to an increase in construction and manufacturing employment. Finally, we show that the impact of wind energy installation on local economic development varies significantly by installed capacity and by county urban/rural status.

Keywords: wind energy, economic development, economic impact, renewable energy

I. Introduction

In the past 25 years, wind energy has grown from a novel energy source to now representing close to 40 percent of all new commercial energy installations in the United States. Installed wind energy capacity has grown from less than 2 gigawatts in 1995 to over 122 gigawatts in 2020 (Center for Sustainable Systems, 2021). The majority of these wind farms are located in rural communities and are increasingly viewed as a potential source of economic development for rural counties which have struggled to recover since the Great Recession (AWEA 2020; Munday, Bristow and Cowell, 2011; Shoeib et al., 2021).

While there is a robust literature on the impact of wind energy on property values (Hoen et al., 2011; Lang et al., 2014; Hoen et al., 2015; Gibbons, 2015; Hoen and Atkinson-Palombo, 2016; Sunak and Madlener, 2016; Jensen et al., 2018), a relatively small set of studies have examined the impact of commercial wind energy on local economic development. What evidence is available comes primarily from studies that focus on specific states or regions and use project-based case studies or input/output models to estimate the effect of wind energy on local economic development (e.g., De Silva et al., 2016; Slattery et al., 2011; Loomis et al., 2016; Faturay et al., 2020). Furthermore, much of this literature is descriptive, providing estimates of the conditional correlation rather than the causal relationship, between wind energy installation and local economic development.¹

The purpose of this paper is to provide new and nationally representative evidence on the causal impact of commercial wind energy installation on local economic development. We exploit variation in the location and timing of adoption of wind energy installations across counties to provide new evidence on the causal impact of commercial wind energy on local economic development. We combine data on the timing, location, and capacity of the universe of wind energy installations in the U.S. from 1995-2018

¹ Exceptions include Mauritzen (2020) who uses a difference-in-difference identification strategy to examine the impact of wind energy installation on county wages and Shoeib, Renski and Infield (2022) who use difference-in-differences and propensity score matching methods to examine the impact of large wind energy installations in rural U.S. counties. We discuss these papers in more detail in Section II.

with data from the Bureau of Economic Analysis (BEA), U.S. Bureau of Labor Statistics, and the U.S. Census on a host of annual county-level measures related to economic development, namely: 1) GDP per-capita; 2) income per-capita; 3) median household income; 4) median home values; 5) total employment; and 6) the share of employment by industry.

To isolate the causal effect of wind energy installation on county-level economic development, we first estimate non-parametric event study models that allow us to examine the evolution of our outcomes both before and after the introduction of commercial wind energy in a county. We then estimate difference-in-differences instrumental variables models (DD IV) where we relate our outcomes of interest to annual installed wind energy capacity per-capita (measured in megawatts) in a county and account for the potentially nonrandom location of wind energy installations by instrumenting for wind energy capacity using county wind speeds. Further, we allow for separate treatment effects of wind energy installation on our outcomes of interest in the construction phase (i.e., two years prior to an installation generating electricity) and the operation phase (after an installation starts generating electricity).

We find that wind energy installation led to economically meaningful and statistically significant increases in county-level GDP per-capita, income per-capita, median household income, and median home values. The increases in GDP, per-capita income, and median income begin in the construction phase and accelerate during the operation phase. In contrast, wind energy installation appears to affect housing values only during the operation phase. While we find that wind energy had little impact on total employment, wind energy installation changes the composition of the local labor market with a shift away from farm employment towards non-farm employment, notably through an increase in manufacturing and construction employment.

We also document significant heterogeneity in all of our outcomes based on installed wind energy capacity per-capita. Specifically, counties with installed wind capacity of less than the median experienced only small changes in our outcomes of interest while those above the median and particularly, those above the 75th percentile of installed capacity experienced much larger effects. Finally, we find that the impact of wind energy installation differed for rural and non-rural counties, with rural

counties experiencing larger increases in GDP per-capita and the share of employment in the construction and manufacturing sectors. We also find that while the adoption of commercial wind energy led to declines in total population in more urban counties it had no effect on total population in rural counties.

2. Background and Related Literature

Wind energy is an increasingly important energy source in the United States, particularly in the western and southwestern states. In 2020, there were over 1,500 commercial wind installations in the United States, consisting of over 60,000 utility-scale wind turbines capable of producing enough electricity to power 32 million homes (AWEA 2020). Over the past three decades, both the number of commercial wind installations and installed wind energy generating capacity has grown tremendously. Figure 1 presents the distribution of county-level wind energy capacity in 1995 and 2019. As shown in Figure 1A, in 1995, wind energy production was extremely rare and was concentrated almost entirely in California and to a lesser degree in Texas and there were only 7 counties in the U.S. with wind energy installations, producing less than 2 gigawatts of power. As shown in Figure 1B, by 2019, installed wind energy generating capacity had grown to over 122 gigawatts and wind energy production had spread across 670 counties and 38 states. The vast majority of these wind energy installations are concentrated in the western, south western, and central United States, and there is significant variation in installed wind energy capacity both across states, as well as across counties within a state.

As noted by Brown et al. (2012), wind energy advocates often tout the potential local economic benefits, as well as the global, national, and local environmental benefits, of wind energy. There are a number of ways that commercial wind energy installations can affect the local economies in which they are situated. Commercial wind energy can have a *direct effect* on local employment, income and GDP per-capita by providing jobs, notably construction jobs, to those working on building the wind installations and income through lease payments to local landowners. These income and employment effects should materialize during the construction phase, as work begins on constructing the turbines and lease payments begin to landowners, and continue through the operation phase, although the employment

effects during the operation phase may be diminished as less labor is needed once the turbines become operational. Commercial wind energy can also have *indirect effects* on local economic development via the effect of wind installations (and those working on building the installations) on the consumption of locally produced goods (e.g., hardware, constructional material, etc.) and services (e.g., restaurants, hotels, environmental management, consultants, etc.). These indirect consumption effects will provide additional income and increase employment for these other services and sectors in the local economy.

Commercial wind energy installations can also directly affect the local tax base of the jurisdictions in which they reside. The increase in the local tax base along with any payments in lieu of taxes (PILOTS) or other forms of indirect compensation to local governments, can then be used by local governments to increase public services and/or provide property tax relief to local citizens. Consistent with that notion, Kahn (2013), Silva et al. (2016), Castleberry and Green (2017), and Brunner, Hoen, and Hyman (forthcoming) find evidence that wind energy installation led to increases in public school spending and in some cases declines in local property tax rates. In related work, Brunner and Schwegman (2021) find evidence that wind energy installation led to increases in county own-revenues and expenditures with particularly large increases in highway and hospital expenditures.

As noted previously, much of the existing literature on the impact of wind energy installation on economic development focuses on regional or national impacts and uses project-based case studies or input/output models to estimate the economic impacts of wind power developments (e.g., Government Accountability Office, 2004; Greene and Geisken, 2013; Pedden, 2006; Slattery et al., 2011; Loomis et al., 2016; Faturay et al., 2020). More recently a smaller set of studies has begun to use econometric techniques to estimate the economic impact of wind power developments at the local level (e.g., Brown et al., 2012; Hartley et al., 2015; De Silva et al., 2016; Mauritzen, 2020; Shoeib et al., 2022).

Using an ex-post econometric model and national data on changes in installed wind energy capacity per-capita from 2000 through 2008, Brown et al. (2012) found an \$11,000 increase in per-capita income and an increase of 0.5 jobs per megawatt of wind power capacity per-capita. Using county-level data from Texas, Hartley et al. (2015) and De Silva, McComb, and Schiller (2016) find a fairly small

impact of wind energy installation on local employment and wages, while De Silva, McComb, and Schiller (2016) find significant increases in per capita income accompanying wind power development. Most recently, Shoeib et al. (2021) examine the impact of large (over 100 MW in 2012) wind energy installations on community services, the cost of living and employment in 11 rural counties located in five different states. They provide descriptive evidence that wind energy installation led to a modest increase in construction employment (particularly during the construction phase) and that wind development tax income improved community services without any noticeable increase in local taxes or cost of living.

In the two studies closest to ours, Mauritzen (2020) uses national data on rural counties to examine the impact of wind energy installation on county wages while Shoeib et al. (2022) uses data on rural and non-rural counties to examine the impact of wind energy installation on employment, income, poverty rates and population. Using a Bayesian multilevel model, and a difference-in-differences research design, Mauritzen (2020) finds evidence of a modest increase in wages associated with wind energy. Similarly, using a difference-in-differences research design and data on wind energy installations from 1990 to 2015, Shoeib et al. (2022) find that wind energy led to increases in per-capita income and small increases in employment per-capita. They also find that wind energy installation is associated with small declines in county-level poverty rates, unemployment rates and population, particularly for counties with higher wind speeds.²

We contribute to this existing literature in several ways. First, we employ an identification strategy that explicitly accounts for biases that can arise due to nonrandom selection into treatment (i.e., adoption of wind energy installations) and in models with staggered timing of treatment and heterogeneous treatment effects. Second, we provide new evidence on the heterogeneous effects of wind energy installations with respect to wind energy capacity per-capita and with respect to whether a wind energy installation is located in a more rural or urban county. Third, we examine the effect of wind energy

² Using propensity score matching techniques, Shoeib et al. (2022) also find that wind energy installation had insignificant effects on county employment per-capita, unemployment rates or per-capita income in rural counties with large wind energy installations (over 100 MW).

installations on a broader set of outcomes including median home values and manufacturing and construction employment shares within a county.

Finally, our paper contributes to the literature on how wind energy affects home values. A relatively large body of literature has examined how proximity to wind turbines impacts individual home values. Within the U.S. context, the evidence is mixed but the majority of the literature has found insignificant effects of wind turbines on home values (e.g., Laposa and Mueller 2010; Hoen et al. 2015; Sampson et al. 2020). For example, Hoen et al. (2015) use data on over 50,000 homes, including 1,198 homes within 1 mile of a turbine and find negative but small and insignificant effects of turbines on homes located in close proximity to turbines. Lang, Opaluch & Sfinarolakis (2014), reach a similar conclusion using data on 48,554 housing transactions in Rhode Island while Sampson et al. (2020) find that wind turbines did not affect agricultural property values. Our paper contributes to this literature by providing some of the first evidence on how large-scale wind energy developments affect home values at the county level via their effect on local economic development.³

3. Data

We construct a county-level panel dataset that combines information on the universe of wind energy installations in the United States with data from a variety of sources pertaining to local economic development. National data on installed wind capacity comes from the United States Wind Turbine Database (USWTDB), which is a turbine-level dataset with detailed information on the date each wind turbine became operational, the nameplate capacity of each turbine measured in kilowatts, and the longitude and latitude of each turbine. These data also include the county in which the turbine is located. We then create a panel dataset containing annual data on total installed wind capacity in each county by aggregating the capacity of every turbine in operation in a county in a given year up to the county level.

³ Brunner and Schwegman (2021) also examine the impact of wind energy installation on county-level home values and find that wind energy increased home values on average by approximately 4 percent. We build on their work by examining heterogeneity in housing value effects by installed capacity per-capita and by whether a county is classified as rural or urban.

We combine these data with county-specific data on Gross Domestic Product (GDP), per-capita income and county-level employment by SIC industry codes from the Bureau of Economic Analysis. The BEA's data on GDP for counties begins in 2001, while data on total employment, farm employment, construction employment, and manufacturing employment is available back to the first year of our sample time frame, namely 1995. We then combine these data with county-level total population, and median household income data from the U.S. Census Bureau (adjusted to real 2012 dollars using the consumer price index).

As we demonstrate later in the paper, wind energy installation led to meaningful declines in county population, particularly in less rural counties. As a result, if we standardize income per-capita, GDP per-capita and other measures of economic development by annual population, we are likely to conflate changes in these measures with changes that occur simply because of the effect wind energy installation has on county population. We therefore standardize all of our outcomes that are measured in per-capita terms by dividing by population in our base year, namely 1995.

Finally, we combine our county panel dataset with data on housing transactions and sale prices from the Federal Housing Finance Agency (FHFA) Housing Price Index (HPI), which provides a county-level weighted repeat-sales HPI for single-family homes with the base year set to 2000. We merge the HPI data with data on median home values for each county from the 2000 decennial census and then inflate this median home value using the HPI. This provides us with the median home value for each county in each year. In keeping with the hedonics literature, in all analyses, we use the log of this home value measure.

In all our specifications we include a number of county-level control variables taken from the 1990 Decennial Census, namely: 1) county population; 2) percent urban; 3) poverty rate; 4) the share of the population 25 or older with a Bachelor's degree or higher; 5) the share of the population that is non-

white; 6) the share of the population that is 65 or older; and 7) the fraction of homeowners. We measure county population in our base year of 1995, while all other variables are from 1990.⁴

We restrict our analytic sample in two primary ways. First, we drop counties that have a cumulative installed wind energy capacity of 2 megawatts (MW) or less.⁵ Second, we drop a small number of counties that had wind installations constructed prior to 1995, the beginning of our time period. In some of our specifications, we also explicitly focus on rural counties to examine whether there are important differences between how wind energy installation affects rural versus non-rural counties. Specifically, we use the 1993 USDA rural–urban continuum codes to identify counties which the USDA categorizes as “rural” implying a rural–urban continuum code of 6 or higher.⁶

Our final sample consists of 2,971 counties (2,000 rural counties) over the period 1995–2018. 465 counties (338 rural counties) had a wind energy installation at some point between 1995 and 2018. Table 1 provides summary statistics for our primary outcomes (top panel) and control variables (bottom panel). We present separate summary statistics for the full sample of counties, and counties with and without wind energy installations in Columns 1 through 3 and then the same information for the sample of rural counties in Columns 4 through 6. To better compare counties with and without commercial wind energy installations, we present summary statistics for our outcomes of interest measured in the first year the data is available. As Table 1 reveals, rural counties tend to have significantly smaller populations, lower income and a higher share of employment in the farming sector. For both the full sample and the

⁴ Given that all our empirical specifications include county fixed effects, all of the above variables enter our specifications as interactions between the variable and a linear time trend to allow for differential trending based on these factors. Also note that all our results are robust if we include no controls or a more parsimonious set of controls as one would expect if commercial wind energy installations are as good as randomly assigned.

⁵ Median turbine nameplate capacity was approximately 2.0 MW in 2016. Thus, a cumulative installed capacity of 2 MW or less, represents a single wind turbine which is typically intended for private on-site use rather than being part of a larger wind energy project designed for commercial electrical generation. Our results are robust to this restriction.

⁶ A code of 6 or higher includes: “Urban, adjacent to metropolitan area, 2,500–19,999 population” (code 6); “Urban, not adjacent to metropolitan area, 2,500–19,999 population” (code 7); “Rural, adjacent to metropolitan area, less than 2,500 population” (code 8); and “Rural, not adjacent to metropolitan area, less than 2,500 population” (code 9). We find similar, but much noisier, results if we define the “rural” sample as 7 or greater.

rural sample, counties with commercial wind installations also tend to have smaller populations, higher shares of the population that is non-Hispanic white, and a higher share of farm employment.

4. Methods

To examine the effect of wind energy on county economic development, we begin by presenting a non-parametric difference-in-differences event study of the following form:

$$y_{it} = \sum_{j=-8}^8 \gamma_j T_{j,it} + \mathbf{X}_i \theta_t \kappa + \delta_i + \lambda_t + \eta_{it}, \quad (1)$$

where y_{it} denotes an outcome of interest for county i in year t and $T_{j,it}$ represents a series of lead and lag indicators for when a wind installation becomes operational in county i . We re-center $T_{j,it}$ so that $T_{0,it}$ always equals one in the year the installation became operational. We include an indicator variable for 8 or more years prior to an installation becoming operational ($-T_{8,it}$), a series of indicators from 0 to 7 years prior to an installation becoming operational ($-T_{-7,it}$ to $T_{0,it}$), and a series of indicators for 1 to 8 or more years after becoming operational ($T_{1,it}$ to $T_{8,it}$). The omitted category for our treatment indicators (i.e. the reference year for all estimates) is three years prior to a wind installation becoming operational ($T_{-3,it}$). Our rationale for choosing three years prior to a wind energy installation becoming operational stems from the fact that, on average, wind energy installations take approximately two years to construct, leading to employment effects and royalty and landlord lease payments prior to operations. In that sense, counties with wind energy installations are already “treated” two years prior to becoming operational. We include both county fixed effects (δ_i) and year fixed effects (λ_t), and η_{it} is a random disturbance term. Given that treatment (wind energy installation) occurs at the county level, in all specifications we cluster the standard errors at the county level.

The coefficients of primary interest in equation (1) are the γ_j 's. The estimated coefficients on the lead treatment indicators ($\gamma_{-8}, \dots, \gamma_{-4}$) provide evidence on whether the parallel trends assumption, which underlies all causal claims based on DD models, appears to hold. If wind energy installation induces exogenous increases in county income, GDP, etc., these lead treatment indicators should be small

in magnitude and statistically insignificant. The lagged treatment indicators ($\gamma_0, \dots, \gamma_{+8}$) allow the effect of wind energy installation on our outcomes of interest to grow over time and in a nonparametric way in the post treatment period.

We include a vector of control variables (X_i), namely 1995 population and 1990 share urban, share 65 or older, share homeowner, share non-white, share of the population with a bachelor's degree or higher and the 1990 poverty rate, all interacted with a linear time trend (θ_t) to allow for differential trending by counties with different baseline (pre-treatment) values of these characteristics. We do not include time-varying control variables because they could be affected by the installation of wind energy (i.e., endogenous controls).

Several recent studies have shown that estimates from standard event studies and DD specifications relying on the staggered timing of treatment for identification may be biased in the presence of heterogeneous treatment effects (Callaway and Sant'Anna 2020; Sun and Abraham 2020; Goodman-Bacon 2021). Specifically, in the presence of heterogeneous treatment effects, event study estimates of lags and leads can be biased by the effects from other relative time periods (i.e., effects from early versus later adopters). To address this concern, we employ the estimator developed by Sun and Abraham (2020), which is free from any contamination and bias that may arise in event study models with staggered timing of treatment and heterogeneous treatment effects.

To improve the precision, we complement the event study specification with a binary difference-in-differences (DD) specification. To avoid the potential biases noted above that arise in models with staggered timing of treatment and heterogeneous treatment effects, we follow Cengiz et al. (2019) and Goodman and Bacon (2021) and estimate a stacked DD model. Specifically, we first create a set of datasets that include observations from a cohort of counties where wind energy development becomes operational in the same year and other counties that never had a wind energy development during our sample timeframe. We then append these cohort-specific datasets and estimate models of the following form:

$$y_{itc} = \alpha_1 \text{Construction}_{it} + \alpha_2 \text{Operation}_{it} + X_i \theta_t + \delta_{ic} + \lambda_{tc} + \varepsilon_{itc}, \quad (2)$$

where y_{itc} is an outcome of interest for county i in year t and a member of cohort c , Construction_{it} is an indicator that takes the value of one in the two years prior to a wind energy installation becoming operational, zero otherwise, Operation_{it} is an indicator that takes the value of one in all years after an installation becomes operational, δ_{ic} and λ_{tc} are vectors of county-by-cohort and year-by-cohort fixed effects respectively, ε_{itc} is a random disturbance term, and all other terms are as defined in equation (1). The coefficients of primary interest in equation (2) are α_1 and α_2 which represent the DD estimate of the effect of treatment (wind energy installation) on our outcomes of interest during the construction phase and the operation phase respectively.

Next, to account for the fact that the capacity of wind energy installations varies across counties and across time, in our preferred specification we allow for continuous treatment by replacing the two binary treatment indicators in equation (2) with the installed wind capacity per-capita in a county in a given year:

$$y_{itc} = \beta_1 \text{Construction_MW}_{it} + \beta_2 \text{Operation_MW}_{it} + X_i \theta_t + \delta_{ic} + \lambda_{tc} + v_{itc}, \quad (3)$$

where MW_{it} is installed wind capacity per-capita in county i in year t measured in megawatts per-capita, v_{itc} is a random disturbance term, and all other terms are as defined in Equation (2).⁷ MW_{it} is equal to zero for county-years with no installed wind energy. Thus, Operation_MW_{it} , equals installed capacity per capita in all years following a wind energy installation becoming operational and zero otherwise and $\text{Construction_MW}_{it}$ equals installed capacity per capita in the two years prior to a wind installation becoming operational and zero otherwise. The coefficients of primary interest in (3) are β_1 and β_2 which represent the effect of a one-megawatt per-capita increase in installed wind energy capacity on our outcomes of interest in the construction and operation phases respectively.

⁷ Similar to our per-capita outcomes, we standardize wind energy capacity by dividing it by 1995 population so that the measure is not affected by changes in population that arise due to treatment.

A potential concern with our DD specifications, is that counties with commercial wind energy installations may be different from other counties along unobservable dimensions that may also affect local economic development (i.e., nonrandom self-selection into treatment). For example, counties that actively pursue commercial wind energy (potentially by offering tax abatements and other incentives) may be better primed and positioned for economic development than other counties, leading to estimates that overstate the impact of commercial wind energy on local economic development. To address that concern, in all our DD models, we account for the possibility of nonrandom self-selection into treatment by adopting a difference-in-differences instrumental variables (DD IV) identification strategy. Specifically, we use average county wind speed as an instrumental variable in our DD framework given by equations (2) and (3).⁸ Note that in our DD IV models, the coefficients of primary interest, β_1 and β_2 , are now identified solely based on the portion of the variation in wind energy capacity per-capita that is explained by wind speeds and thus our IV estimates net out all strategic behavior on the part of counties or other unobservables that might be correlated with economic development.⁹

5. Results

We begin by presenting the impact of commercial wind energy on our main outcomes of interest by plotting the estimated γ_j 's and associated 90 and 95 percent confidence intervals from our event study specification given by Equation 1. Figure 2A, 2B, 2C and 2D show that GDP per-capita, per-capita income, median household income, and median home values (all measured in logs) begin to increase after a county installs wind energy. Importantly, there is no evidence of differential trending prior to the installation of wind energy—the estimated pre-treatment effects are small in magnitude and nearly all are

⁸ Average county wind speed satisfies the conditions for a valid instrument given: 1) it is highly correlated with wind energy installation and total installed capacity (first-stage F-statistic always greater than 100), (2) it is as good as randomly assigned (i.e., counties do not choose their wind speed), and (3) the only reason wind speed should affect county economic development outcomes is through its effect on wind energy adoption and capacity.

⁹To construct the instrument, we interact average county wind speed with an indicator for having installed turbines and use this as an instrumental variable for turbine capacity. Data on county wind speeds come from the Wind Integration National Dataset (WIND) Toolkit (Draxl et al., 2015) and reflects wind speeds at a 100-meter height (typical wind turbine height) over the period 2007-2013.

statistically insignificant. The lack of differential trending prior to the installation of wind energy provides evidence that our main identification assumption—the parallel trends assumption—holds.

We next examine the impact of wind energy installation on county employment patterns. Figure 2E shows that total employment is relatively unaffected by wind energy installation. Figures 2F and 2G show that the share of employment in the construction and manufacturing sectors respectively begin to increase during the construction phase of wind energy installation and continues through the operation phase while Figure 2H shows that these shifts in the share of employment dedicated to construction and manufacturing are accompanied by a small decline in farm employment that begins later in the operation phase. Finally, Figure 2I shows that wind energy installation is associated with a decline in total population relative to counties without wind energy with the decline in population starting during the construction phase and continuing into the operation phase.

We present the DD IV estimates of the impact of wind energy installation in Table 2. Results based on equation (2) with binary treatment are presented in columns 1 and 2 while results based on equation (3) with continuous treatment are presented in columns 3 and 4. All results reported in Table 2 and subsequent tables are DD IV estimates where we use average county wind speed as an instrumental variable for wind energy adoption and wind energy capacity per-capita. Furthermore, similar to our event study models, all of our outcomes other than employment shares and total population are measured in natural logs so that the estimates can be interpreted as rates of return. Column 1 presents the results from equation (2) with only a single treatment indicator that equals one in all years after a wind energy installation become operational (i.e., equation (2) without the $Construction_{it}$ indicator), whereas Column 2 presents the results from the complete version of equation (2).

Beginning with Columns 1 and 2 in the top panel of Table 2, we find that wind energy led to large increases in GDP per capita (approximately 6 percent), income per capita (5 percent), median household income (3 percent), and home values (2.6 percent). As shown in column 2, the impact of wind energy installation on GDP and both income measures begins during the construction phase and the impact increases during the operating phase. For example, GDP per-capita increases by 2 percent during

the construction phase and by 7 percent during the operation phase. In contrast, home values only increase after a wind energy installation has begun operating. Finally, consistent with the event study figures presented above, we find that wind energy installation leads to a decline in total population. This population decline begins during the construction phase (although the estimate is not statistically significant) and accelerates during the operation phase.

We present the employment outcomes in the bottom panel of Table 2. Consistent with the event studies presented above, wind energy appears to have little impact on total employment. However, wind energy installation causes a small decline in farm employment (a decrease of 0.2 percent) and a shift from farm employment towards construction and manufacturing employment. Specifically, wind energy installation leads to a 1.3 percent increase in the share of manufacturing employment and a 0.7% increase in the share of construction employment. As shown in Column 2, this increase in the share of employment in construction and manufacturing began during the construction phase, which is consistent with the event studies presented above.

Our per-capita income results are consistent with Brown et al. (2012), De Silva et al. (2016) and Shoeib et al. (2022) who also find positive impacts of wind energy installation on per-capita income. Similarly, our total employment results are generally consistent with the prior literature which tends to find small effects of wind energy installation on employment.

The DD models with binary treatment are useful because they provide an intuitive and easy to interpret estimate of the impact of wind energy installation on our outcomes. However, these models suffer from two drawbacks. First, as in the event-study analysis, the binary treatment variable turns on when the first installation in a county occurs, and so it does not capture the increased capacity of subsequent installations for the 58.71 percent of counties with multiple installations over time. Second, the binary treatment variable misses important variation stemming from different wind energy installations having very different installed capacity. The distribution of installed wind capacity per capita is heavily right skewed: the 10th and 25th percentiles of installed capacity per capita in our sample among

counties with installed wind energy are 0.0006 MW/capita and 0.0009 MW/capita, respectively, while the 75th and 90th percentiles are 0.014 and 0.036 MW/capita.

Given these limitations, in columns 3 and 4 of Table 2 we present results based on equation (3) where we use a continuous measure of treatment, namely installed capacity (measured in megawatts) per capita. The direct interpretation of these coefficients is the estimated effect of a one MW increase in installed capacity per capita. Thus, an increase of one MW per capita of installed capacity increased GDP per capita by 507 percent. Note however, that the average installed capacity in the full sample is 0.0168 MW/capita. Thus, as we show in column 5, evaluated at the mean capacity per-capita, wind energy installation increases GDP per capita by 8.5 percent ($5.07 \text{ MW} / \text{per capita} \times 0.0168$). Similarly, we find that the average wind installation increased per capita income by approximately 6 percent, median household income by 4 percent, home values by approximately 7 percent, and decreased total population by approximately 5,300 people. Consistent with column 2, we find that wind energy installation began affecting GDP, per capita income, and median household income during the construction phase (see Column 4).

As seen in the bottom panel of Table 2, we continue to find no impact on total employment: the estimated coefficients are small in magnitude and statistically insignificant. However, we continue to find a decrease in farm employment—a 0.3 percent decline ($-0.18 \times 0.016 \text{ MW} / \text{capita}$) at the mean of capacity per-capita. Using a continuous measure of installed capacity, we find that wind energy led to a 1.6 percent increase in the share of manufacturing employment and a 0.8 percent increase in the share of construction employment.

A. Heterogeneity by Capacity per Capita and Rural and Non-Rural Designation

As noted previously, there is wide variation in the distribution of capacity per capita across counties with wind energy installations. Consequently, the mean effects reported in column 5 of Table 2 could mask important heterogeneity in the impact of wind energy installation across the distribution of capacity. Thus, in columns 6 and 7 of Table 2 we show effects at the median and 75th percentile of capacity per capita. For example, the effect of wind energy installation on GDP per capita at the median is

a 1.8 percent increase (relative to the GDP per-capita in the control counties), and at the 75th percentile the effect is 7.2 percent. Similarly, the effects on home values at the median and 75th percentile are 2 percent (a \$1,886¹⁰ increase) and 6 percent (a \$5,658 increase), respectively.

We show in Figure 3 the effects on a select number of our outcomes of interest by ventiles of installed capacity, revealing the same patterns. The effects of wind installation on our outcomes of interest for counties in the bottom half of the distribution are quite small but increase at an increasing rate past the 50th percentile.

Proponents have touted commercial wind energy as a potential source of economic development for rural counties which have struggled to recover both in terms of job and population growth since the Great Recession. Thus, in Figure 4, we present event study estimates of the impact of wind energy installation on a select number of outcomes separately for urban counties (in blue) and rural counties (in red).¹¹

Figures 4A through 4F reveal some important heterogeneity between rural and non-rural counties. For example, while wind energy has a negative impact on population in urban counties, it has very little, if any, impact on the population of rural counties (see Figure 4F). The impact of wind energy on the local economy (GDP) is much greater in rural counties than urban counties (see Figure 4A) and as shown in Figure 4D and 4E, wind energy installation led to a greater shift towards construction and manufacturing employment in rural counties compared to urban counties. Perhaps more importantly, this shift towards manufacturing and construction employment persists past the construction phase.

Figure 5 provides one explanation for why rural counties appear to benefit more in terms of GDP per-capita and employment than more urban counties. Specifically, Figure 5 shows the distribution of installed wind energy capacity per-capita among rural and more urban counties, with the mean value of capacity per-capita in both distributions illustrated with a vertical line. As the figure makes clear, the

¹⁰ The average home value in counties without turbines in 2000 is \$94,298.

¹¹ Specifically, we present separate estimates based on equation (1) where the full sample of counties is split based on whether it is coded as rural or more urban.

distribution of capacity is significantly more rightward skewed among the sample of rural counties with mean capacity per-capita being significantly higher among rural counties (0.0253 MW per-capita) than more urban counties (0.0017 MW per-capita).

We present DD IV estimates of the impact of wind energy installation separately for rural and more urban counties in Table 3. In the interest of brevity, we present only the results based on our preferred specification with continuous treatment (i.e. equation 3) and show estimated impacts at the mean, 50th and 75th percentiles of the distribution of capacity per-capita among the respective samples of rural and more urban counties. The results reported in Table 3 mirror the event study results presented in Figure 4. Specifically, we find that wind energy installation has little impact on the total population of rural counties. This stands in stark contrast to the results for more urban counties where we find a relatively sharp decline in population. Similar to the event studies above, wind energy installation has a larger impact on GDP, and to a slightly lesser extent, per-capita and median income among rural counties than among more urban counties. Finally, the impact on the composition of the local labor market is more pronounced in the rural sample than the full sample: wind energy installation has a greater impact on the share of the local labor market in the construction and manufacturing sectors in rural counties than in more urban counties.

6. Conclusion & Policy Implications

In this paper, we use data on the timing, location, and capacity of the universe of wind energy installations in the U.S. from 1995 through 2018 to provide novel evidence on the impact of commercial wind energy on county economic development. We use event-study and difference-in-differences instrumental variable methodologies that exploit the plausibly exogenous timing and location of wind energy installations and variation in average county wind speeds to examine the impacts of wind energy installation on county-level GDP per-capita, income per-capita, median household income, median home values, total employment, and the share of employment by industry.

We find that wind energy installation led to exogenous and economically meaningful increases in in county-level GDP per-capita, income per-capita, median household income and median home values

with the increases in county GDP and income beginning in the construction phase of wind energy installation and accelerating during the operation phase. While we find that wind energy installation has little impact on total employment, the composition of local employment shifts away from farm employment towards non-farm employment, notably leading to increases in construction and manufacturing employment. Finally, we show that the impact of wind energy installation on local economic development varies significantly by installed capacity and by county urban/rural status.

To reduce its carbon emissions and transition away from fossil fuels, the United States must invest in renewable energies. Collectively, the results of this paper suggest that wind energy not only provides a renewable source of energy but it also has the potential to improve the local economies of the communities in which these energy installations are located. Furthermore, our results suggest that the positive impacts of commercial wind energy on county economic development are larger for more rural counties. Specifically, relatively to more urban counties with wind energy installations, rural counties experience larger increases in GDP per-capita and the share of employment in construction and manufacturing with no measurable displacement of the population. As rural communities continue to struggle with population decline and a limited tax base, our results suggest that wind energy investments may stimulate and diversify local rural economies.

References

- American Wind Energy Association (AWEA). 2020. Wind Powers America Annual Report 2019.
- Brown, J. P., Pender, J., Wiser, R., Lantz, E., & Hoen, B. (2012). Ex post analysis of economic impacts from wind power development in US counties. *Energy Economics*, 34(6), 1743-1754.
- Brunner, E, Hoen, B., & Hyman J. (forthcoming). School District Revenue Shocks, Resource Allocations, and Student Achievement: Evidence from the Universe of U.S. Wind Energy Installations. *Journal of Public Economics*.
- Brunner, E. & Schwegman, D. (2021). Windfall Revenues from Windfarms: How do County Governments Respond to Increases in the Local Tax Base Induced by Wind Energy Installations?. Working Paper.
- Callaway, B., & Sant'Anna, P. H. (2020). Difference-in-differences with multiple time periods. *Journal of Econometrics*.
- Castleberry, B. & Greene, S.. (2017). Impacts of Wind Power Development on Oklahoma's Public Schools. *Energy, Sustainability and Society* 7(1), 34.
- Center for Sustainable Systems, University of Michigan. 2021. "Wind Energy Factsheet." Pub. No. CSS07-09
- De Silva, D. G., McComb, R. P., & Schiller, A. R. (2016). What blows in with the wind?. *Southern Economic Journal*, 82(3), 826-858.
- Faturay, F., Vunnavu, V. S. G., Lenzen, M., & Singh, S. (2020). Using a new USA multi-region input output (MRIO) model for assessing economic and energy impacts of wind energy expansion in USA. *Applied Energy*, 261, 114141.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*.
- Greene, J. S., & Geisken, M. (2013). Socioeconomic impacts of wind farm development: a case study of Weatherford, Oklahoma. *Energy, Sustainability and Society*, 3(1), 1-9.
- Hartley, P. R., Medlock III, K. B., Temzelides, T., & Zhang, X. (2015). Local employment impact from competing energy sources: Shale gas versus wind generation in Texas. *Energy Economics*, 49, 610-619.
- Hoen, B., Brown, J. P., Jackson, T., Thayer, M. A., Wiser, R., & Cappers, P. (2015). Spatial hedonic analysis of the effects of US wind energy facilities on surrounding property values. *The Journal of Real Estate Finance and Economics*, 51(1), 22-51.
- Jensen, C. U., Panduro, T. E., Lundhede, T. H., Nielsen, A. S. E., Dalsgaard, M., & Thorsen, B. J. (2018). The impact of on-shore and off-shore wind turbine farms on property prices. *Energy Policy*, 116, 50-59.
- Kahn, M. E. (2013). Local non-market quality of life dynamics in new wind farms communities. *Energy Policy*, 59, 800-807.

- Lang, C., Opaluch, J. J., & Sfinarolakis, G. (2014). The windy city: Property value impacts of wind turbines in an urban setting. *Energy Economics*, 44, 413-421.
- Loomis, D. G., Hayden, J., Noll, S., & Payne, J. E. (2016). Economic impact of wind energy development in Illinois. *Journal of Business Valuation and Economic Loss Analysis*, 11(1), 3-23.
- Mauritzen, J. (2020). Will the locals benefit?: The effect of wind power investments on rural wages. *Energy Policy*, 142, 111489.
- Mills, S. (2018). Wind energy and rural community sustainability. In *Handbook of Sustainability and Social Science Research* (pp. 215-225). Springer, Cham.
- Munday, M., Bristow, G., & Cowell, R. (2011). Wind farms in rural areas: How far do community benefits from wind farms represent a local economic development opportunity?. *Journal of Rural Studies*, 27(1), 1-12.
- Sampson, G. S., Perry, E. D., & Tayler, M. R. (2020). The On-Farm and Near-Farm Effects of Wind Turbines on Agricultural Land Values. *Journal of Agricultural and Resource Economics*, 45(3), 410-427.
- Shoeib, E. A. H., Infield, E. H., & Renski, H. C. (2021). Measuring the impacts of wind energy projects on US rural counties' community services and cost of living. *Energy Policy*, 153, 112279.
- Slattery, M. C., Lantz, E., & Johnson, B. L. (2011). State and local economic impacts from wind energy projects: Texas case study. *Energy Policy*, 39(12), 7930-7940.
- Sun, L., & Abraham, S. (2020). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*.

Figure 1: Changes in Installed Wind Energy Capacity over Time

Installed Capacity in 1995



Figure 1B: Installed Capacity in 2019

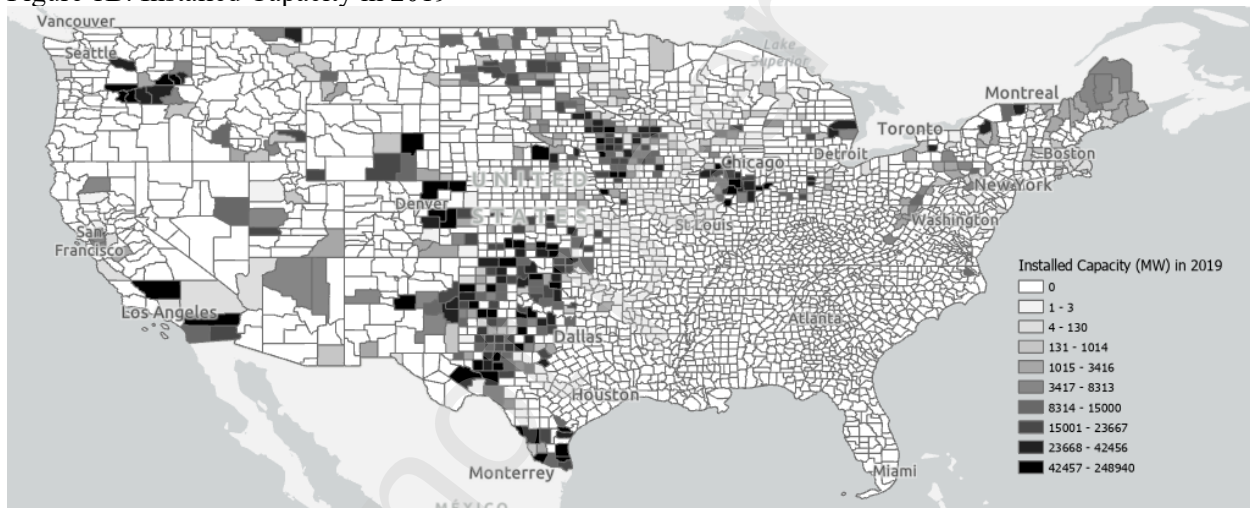
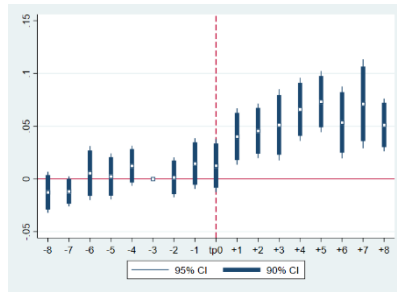
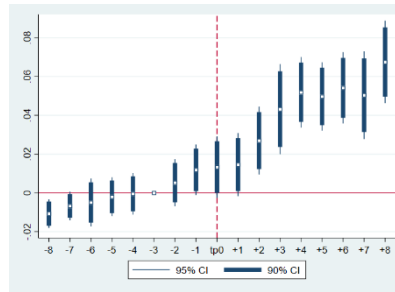


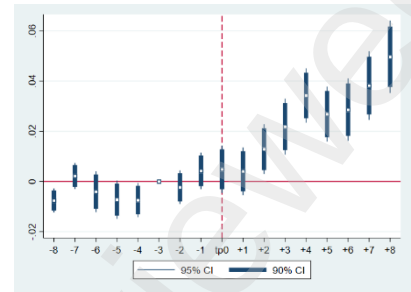
Figure 2: Event Studies on Main Outcomes



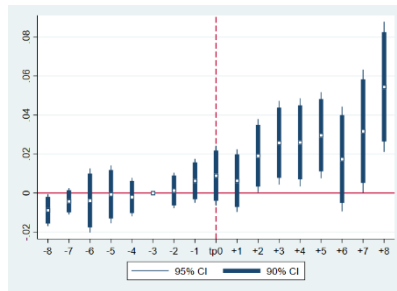
2A: Log of GDP per Capita



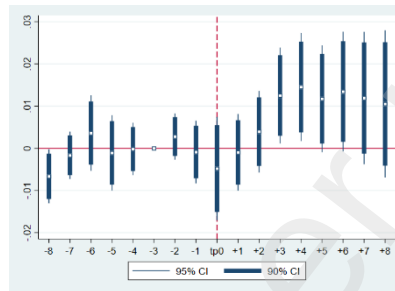
2B: Log of Per Capita Income



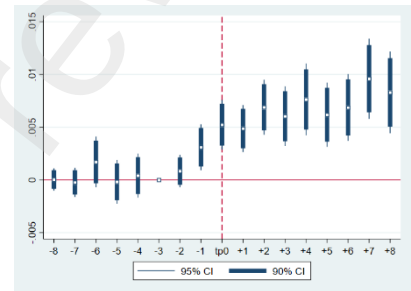
2C: Log of Median Household Income



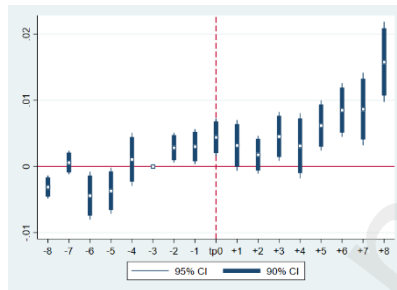
2D: Log of Median Home Value



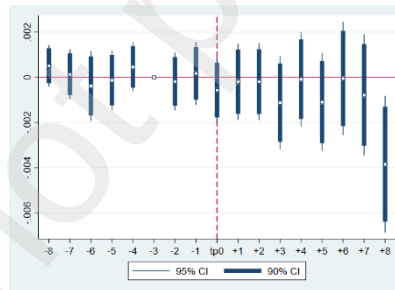
2E: Log of Total Employment



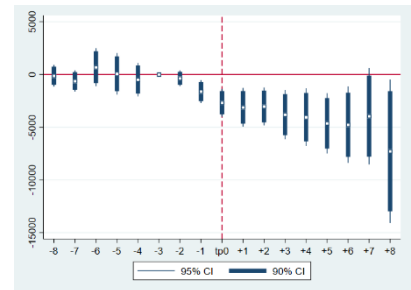
2F: Share of Employment in Construction Sector



2G: Share of Employment in the Manufacturing Sector



2H: Share of Employment in the Farm Sector



2I: Total Population

Figure 3: Impact of Wind Energy Installation on Outcomes of Interest at Various Percentiles of the Distribution of Installed Capacity Per Capita

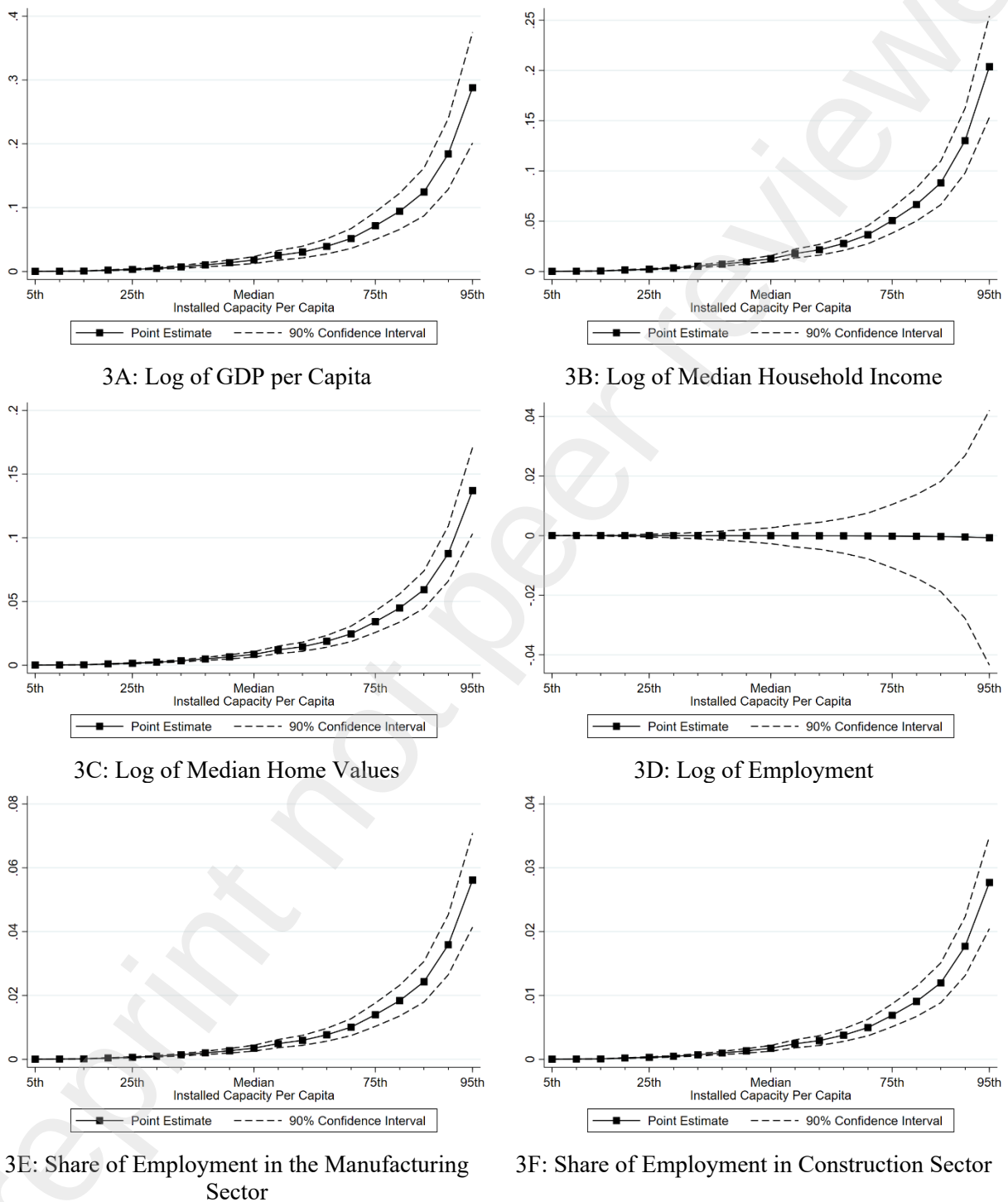


Figure 4: Event Study Graphs Comparing Urban and Rural Counties

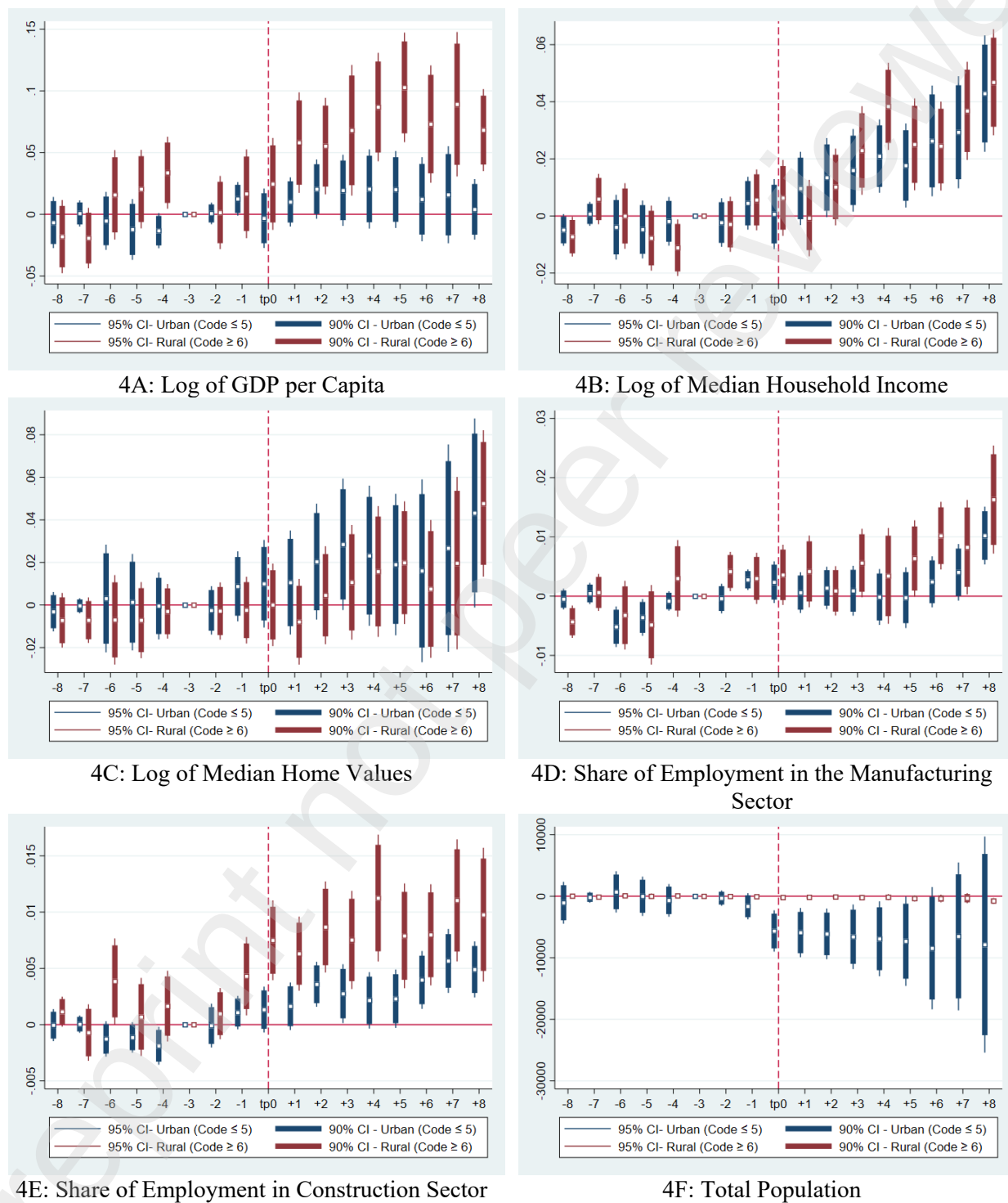


Figure 5: Distribution of Installed Wind Energy Capacity Per-Capita among Rural and More Urban Counties

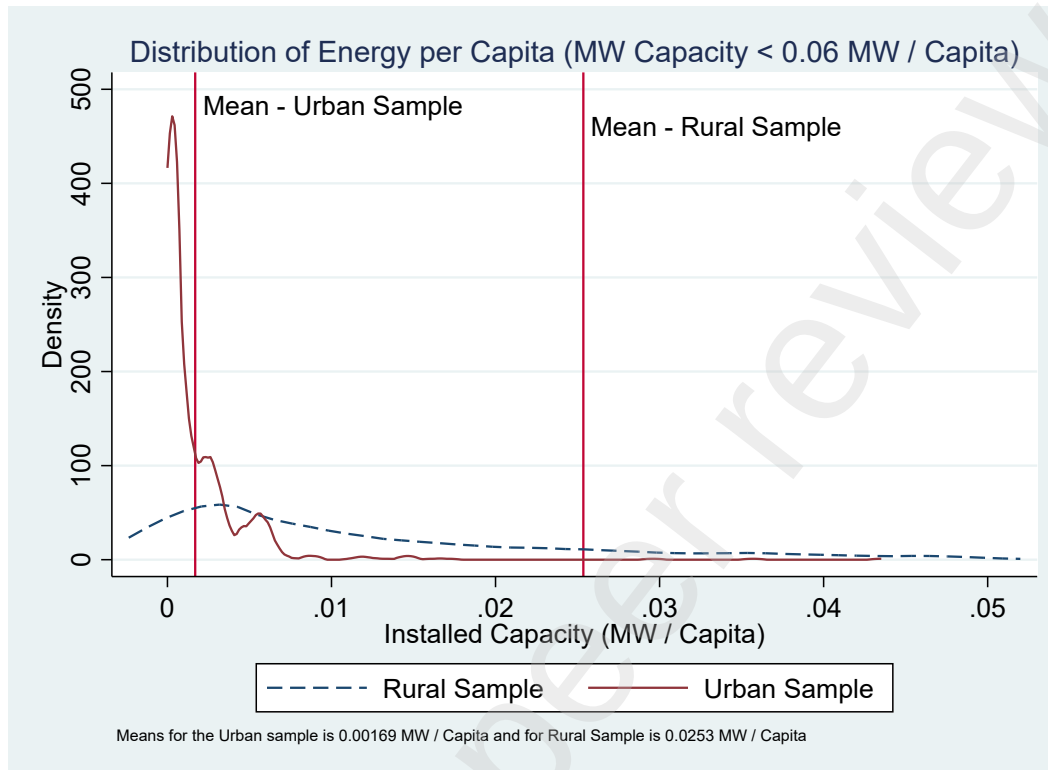


Table 1: Descriptive Statistics

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Sample	Counties with Turbines	Counties without Turbines	Rural Counties	Rural Counties with Turbines	Rural Sample without Turbines
Socio-Economic Covariates						
Population (1990)	131,125	107,396	135,462	21,680	18,383	22,413
Share Non-White (1990)	0.14	0.10	0.15	0.13	0.08	0.14
Share Homeowner (1990)	0.72	0.71	0.72	0.74	0.74	0.74
Share 65+ Population (1990)	0.19	0.20	0.18	0.20	0.22	0.20
Share BA or Higher (1990)	0.14	0.14	0.14	0.11	0.12	0.11
Poverty Rate (1990)	0.15	0.15	0.15	0.18	0.15	0.18
Percent Urban (1990)	0.46	0.46	0.46	0.27	0.31	0.26
Outcomes						
Per Capita Income (1990)	\$ 27,694	\$ 26,820	\$ 27,852	\$ 24,847	\$ 25,390	\$ 24,726
Median Household Income (1990)	\$ 32,336	\$ 31,523	\$ 32,481	\$ 28,369	\$ 29,442	\$ 28,131
GDP Per Capita (2000)	38.58	39.77	38.38	36.18	40.03	35.15
Log of Home Values (2001)	11.36	11.27	11.38	11.22	11.17	11.23
Total Employment (1990)	72,934	53,393	76,498	11,139	9,718	11,455
Share Farm Employment (1990)	0.07	0.10	0.06	0.11	0.15	0.10
Share Manufacturing Employment (1990)	0.15	0.11	0.16	0.15	0.10	0.16
Share Construction Employment (1990)	0.06	0.05	0.06	0.05	0.05	0.05
N of Counties in Sample	2,971	465	2,506	2,000	338	1,662

Table 2: Effect of Wind Energy on the Full Sample

	(1)	(2)		(3)	(4)		(5)	(6)	(7)
	Treatment: Wind Turbine Installed	Treatment: Wind Turbine Installed with Construction Phase Indicator		Treatment: Installed Turbine Capacity Per-Capita (MW)	Treatment: Installed Turbine Capacity Per- Capita (MW) with Construction Phase Indicator		Effect by Capacity Per Capita (MW)		
		Construction Phase	Operating Phase		Constructio n Phase	Operating Phase	Mean	50th pctl	75th Pctl
Non-Employment Outcomes									
Log of GDP Per Capita ¹	0.065*** (0.012)	0.023* (0.012)	0.071*** (0.014)	5.068*** (0.929)	3.106* (1.719)	5.618*** (1.100)	0.085*** (0.016)	0.018*** (0.003)	0.072*** (0.013)
Log of Income Per Capita ¹	0.049*** (0.006)	0.030*** (0.006)	0.054*** (0.007)	3.586*** (0.540)	3.912*** (0.867)	3.993*** (0.596)	0.060*** (0.009)	0.013*** (0.002)	0.051*** (0.008)
Log of Median Household Income	0.033*** (0.004)	0.018*** (0.004)	0.036*** (0.005)	2.412*** (0.363)	2.352*** (0.604)	2.657*** (0.403)	0.041*** (0.006)	0.009*** (0.001)	0.034*** (0.005)
Log of Home Values	0.026*** (0.009)	0.012 (0.009)	0.028*** (0.009)	4.229*** (1.459)	3.905 (2.743)	4.615*** (1.634)	0.071*** (0.024)	0.015*** (0.005)	0.060*** (0.021)
Total Population	-4,333** (1,695)	-1,065 (1,223)	-4,503** (1,860)	-318,055** (131,248)	-139,864 (162,321)	-332,600** (145,181)	-5352** (2209)	-1122.48** (463)	-4499** (1857)
Employment Outcomes									
Log of Total Employment per Capita ¹	-0.000 (0.006)	0.006 (0.007)	0.001 (0.007)	-0.013 (0.459)	0.776 (0.907)	0.068 (0.523)	0.000 (0.007)	-0.000 (0.002)	-0.000 (0.006)
Share Farm Employment	-0.002** (0.001)	-0.002** (0.001)	-0.003** (0.001)	-0.177** (0.070)	-0.261** (0.114)	-0.204*** (0.078)	-0.003** (0.001)	-0.001** (0.000)	-0.003** (0.001)
Share Manufacturing Employment	0.013*** (0.002)	0.013*** (0.002)	0.016*** (0.002)	0.988*** (0.158)	1.692*** (0.260)	1.164*** (0.179)	0.016*** (0.003)	0.003 (0.001)	0.014*** (0.002)
Share Construction Employment	0.007*** (0.001)	0.003*** (0.001)	0.007*** (0.001)	0.487*** (0.077)	0.361*** (0.129)	0.525*** (0.083)	0.008*** (0.001)	0.002*** (0.000)	0.007*** (0.001)

Notes: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors clustered at county-level. All models are estimated using two-stage least squared with wind speed as the instrumental variable with cohort-year and cohort-county fixed effects. (1) Denominator is 1995 population. Average installed capacity is 0.016 MW/Capita, which is larger than both the median installed capacity (0.003MW / capita) and 75th percentile (0.014 MW / capita). In Columns 3 and 4, the coefficients should be scaled by the average installed capacity.

Table 3: Separate Effects of Wind Energy in Rural and More Urban Counties

	(1)	(2)	(3)	(4)	(5)	(6)
	Rural Sample			Urban Sample		
	Effect by Capacity Per Capita (MW)			Effect by Capacity Per Capita (MW)		
	Mean	50th pctl	75th Pctl	Mean	50th pctl	75th Pctl
Installed Capacity (MW / Capita)	0.0253	0.0085	0.0230	0.0017	0.0006	0.0023
Non-Employment Outcomes						
Log of GDP Per Capita	0.087*** (0.019)	0.028*** (0.006)	0.080*** (0.018)	0.028* (0.016)	0.010* (0.006)	0.038* (0.022)
Log of Income Per Capita	0.053*** (0.011)	0.017*** (0.003)	0.048*** (0.010)	0.025*** (0.008)	0.009*** (0.003)	0.034*** (0.010)
Log of Median Household Income	0.035*** (0.006)	0.011*** (0.002)	0.032*** (0.006)	0.021*** (0.007)	0.008*** (0.003)	0.029*** (0.009)
Log of Home Values	0.047** (0.023)	0.015** (0.007)	0.043** (0.020)	0.019 (0.015)	0.007 (0.005)	0.025 (0.020)
Total Population	-309 (368)	-100 (118)	-284 (338)	-8,802** (1,610)	3,265** (1,609)	11,977** (5,904)
Employment Outcomes						
Log of Total Employment per Capita	0.009 (0.008)	0.003 (0.003)	0.008 (0.016)	-0.005 (0.010)	-0.002 (0.004)	-0.006 (0.013)
Share Farm Employment	-0.003** (0.02)	-0.001** (0.000)	-0.003** (0.002)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Share Manufacturing Employment	0.015*** (0.003)	0.005*** (0.001)	0.013*** (0.003)	0.007*** (0.002)	0.003*** (0.001)	0.010*** (0.003)
Share Construction Employment	0.008*** (0.002)	0.002*** (0.000)	0.007*** (0.001)	0.005*** (0.001)	0.002*** (0.000)	0.007*** (0.001)

Notes: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors clustered at county-level. All models are estimated using two-stage least squared with wind speed as the instrumental variable with cohort-year and cohort-county fixed effects. (1) Denominator is 1995 population.

Commercial Wind Energy Installations and Local Economic Development: Evidence from U.S. Counties

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December 2021

Abstract

We examine the causal impact of wind energy installation on the local economies of counties in the United States. Using data on the universe of commercial wind energy installations from 1995-2018 and a difference-in-differences instrumental variable identification strategy, we find that wind energy installation led to economically meaningful and statistically significant increases in county GDP per-capita, income per-capita, median household income, and median home values. We also find evidence that while wind energy installation has little effect on total employment, the composition of local employment shifts away from farm towards non-farm employment, notably leading to an increase in construction and manufacturing employment. Finally, we show that the impact of wind energy installation on local economic development varies significantly by installed capacity and by county urban/rural status.

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